

Multi-factor ultrafine particle ($PC_{0.1}$) concentration optimization prediction and its application

ระบบการคาดการณ์ความเข้มข้นของอนุภาคขนาดเล็กพิเศษ

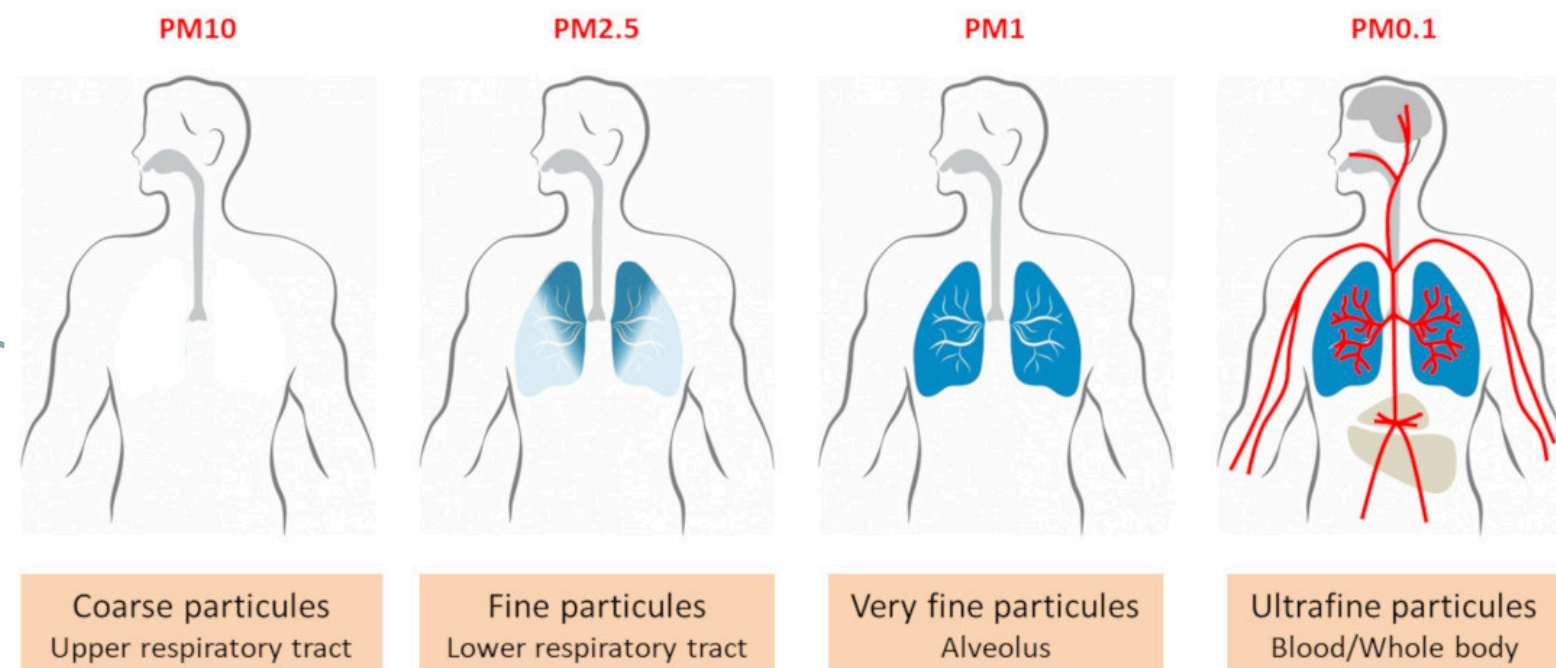
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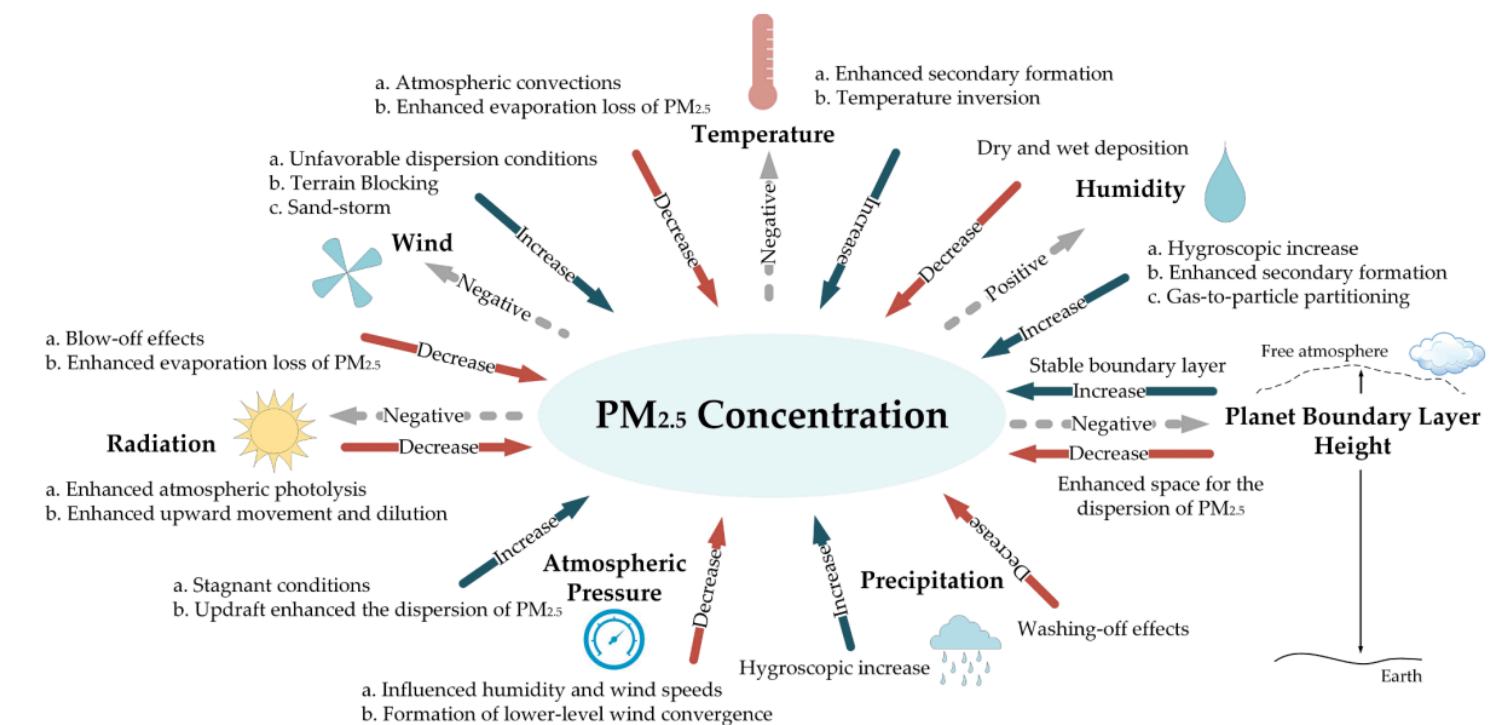
Project Presentation

Importance and Background of the Problem

- Global Air Pollution Impact
- Health Risks of PCO.1
- Public Health and Risk Prevention
- Challenges in Detecting PCO.1



Particle Size



Particle Concentration

Objective

1

To design a conceptual framework

For developing a prototype system for prediction PCO.1 levels.

2

To develop a model capable of predicting

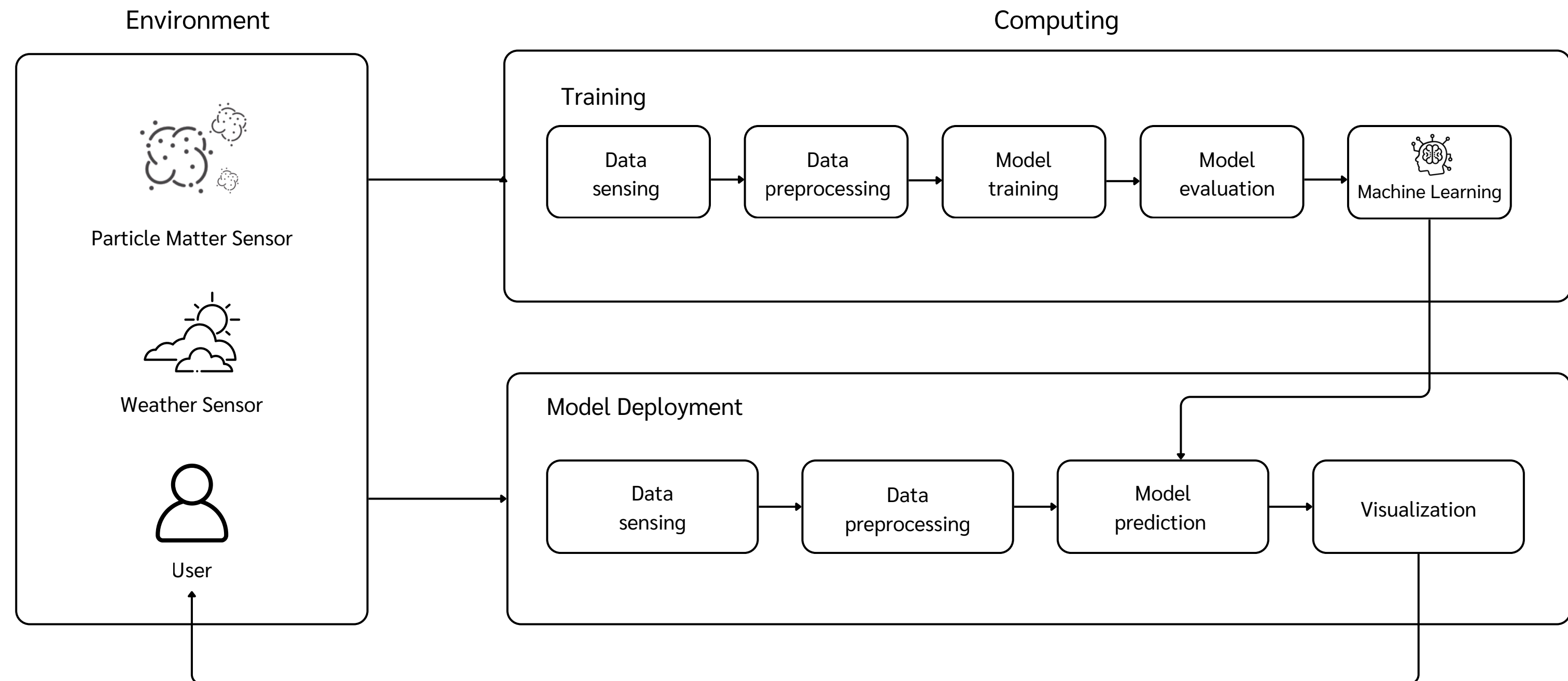
Indoor UFPs concentration by examining the relationships between environmental factors, such as PM2.5, PM10, and indoor factors, with PCO.1 concentration.

3

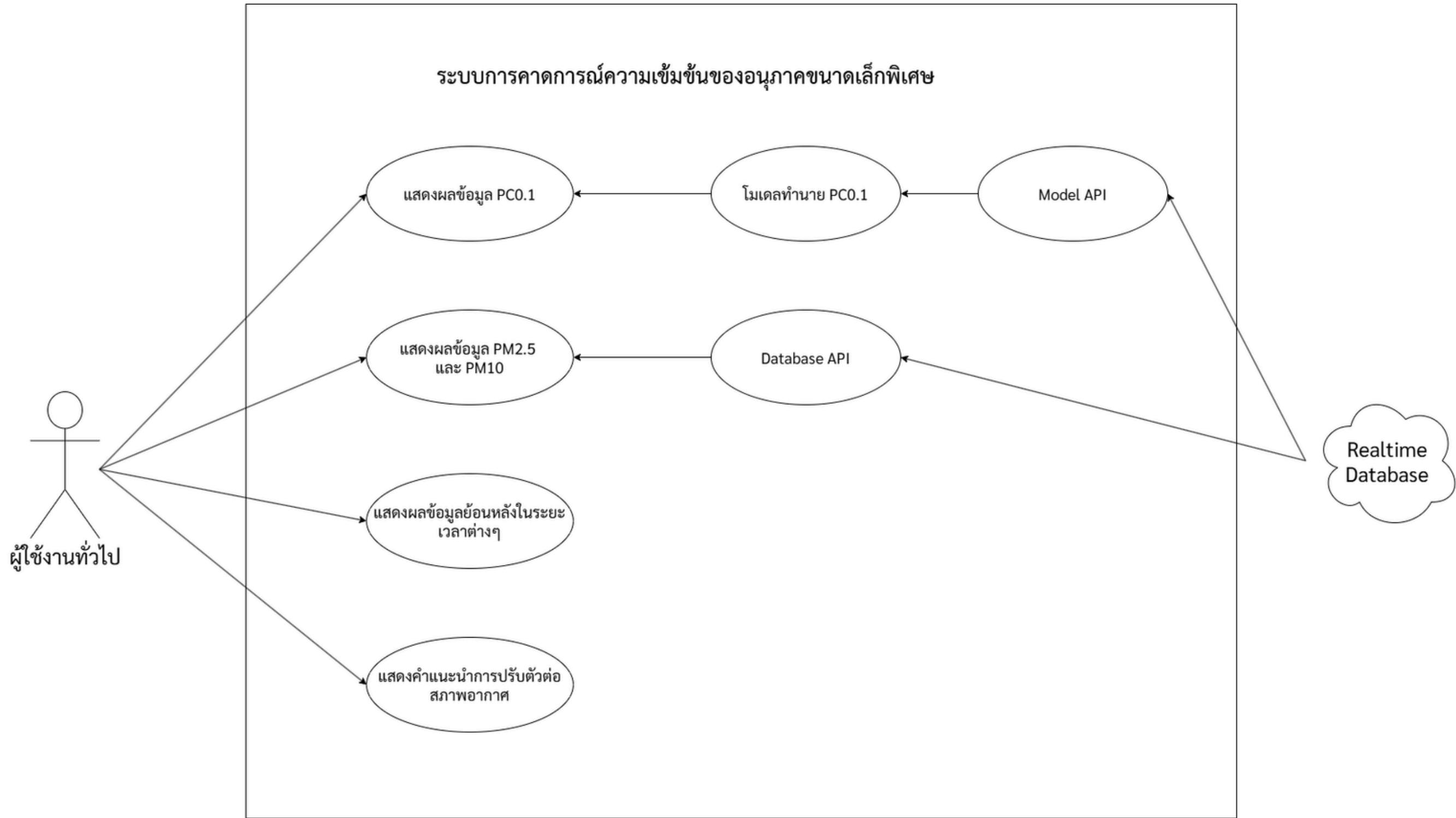
To develop a prototype system

For prediction PCO.1 levels.

Project Framework



Use case



Database Design

RawData	
Location	string
TimeStamp	date/time
PM2.5	float
PM10	float
Humidity	float
Temperature	float

Preprocessed Data	
Location	string
TimeStamp	date/time
PM2.5calibrated	float
PM10	float
Humidity	float
Temperature	float
DayTypes	string
TimeRange	string

Predicted Data	
Location	string
TimeStamp	date/time
Days	string
DayTypes	string
TimeRange	string
PC0.1	float
PM2.5	float
PM10	float
Humidity	float
Temperature	float
Hourly Mean PC0.1	float
Hourly Mean PM2.5	float
Hourly Mean PM10	float
Daily Mean PC0.1	float
Daily Mean PM2.5	float
Daily Mean PM10	float

Sensor comparison



Piera IPS-7100

PM Level: PM0.1 – PM10

Price: 6,000฿

Sampling Time: 0.2 s

Error rate: +- 10 %



Nova PM SDS011

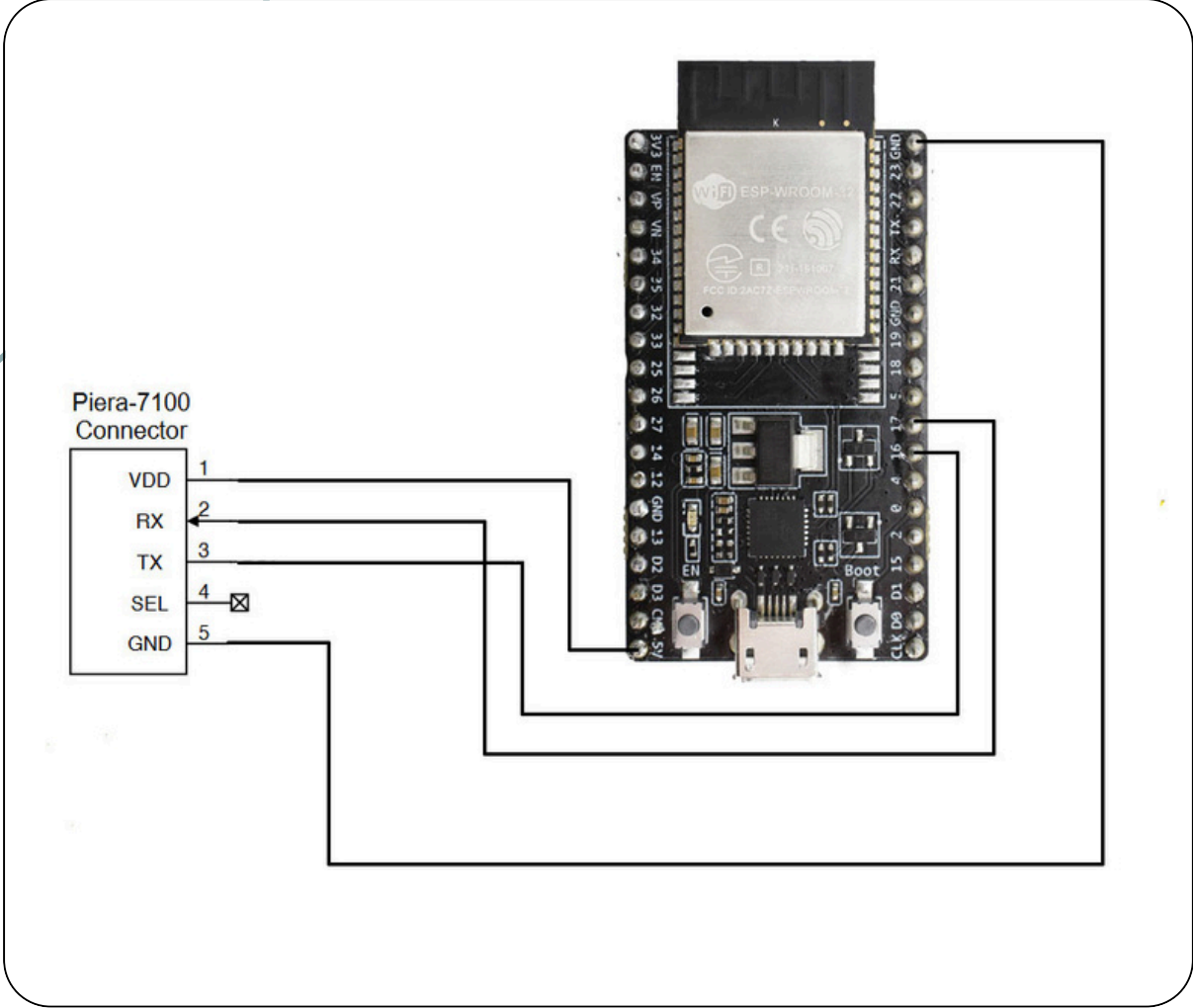
PM Level: PM2.5, PM10

Price: 700 – 1,000฿

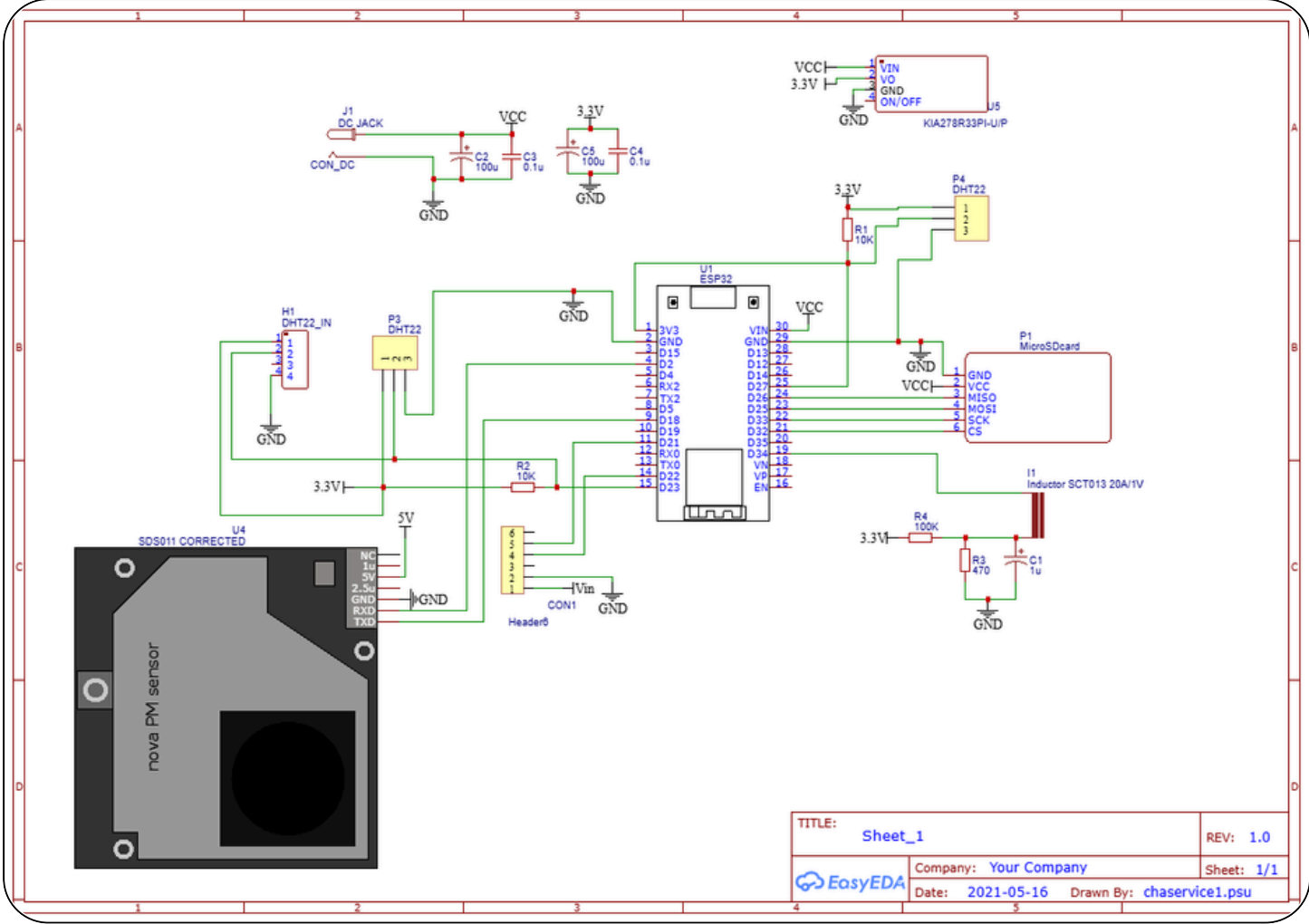
Sampling Time: 1 s

Error rate: +- 15 %

Sensor Setup

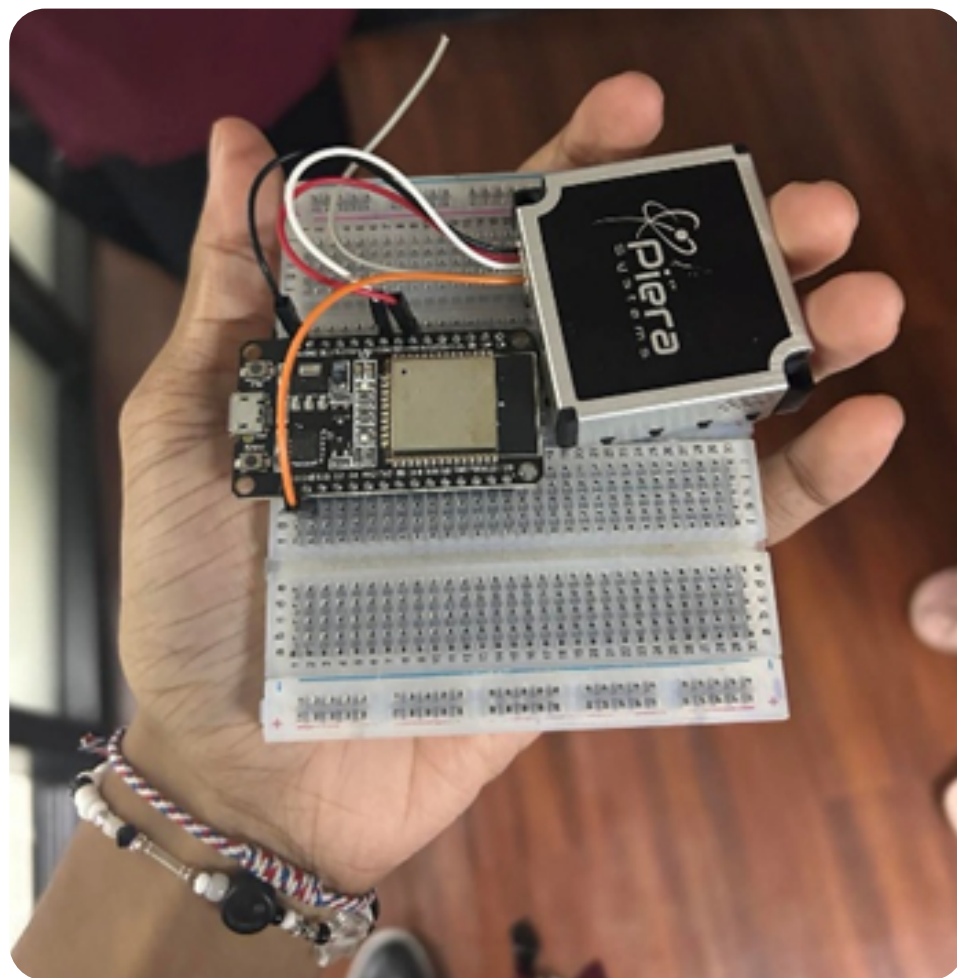


Piera-7100 Setup



Nova PM Setup

Sensor Setup cont.



Piera-7100 Setup

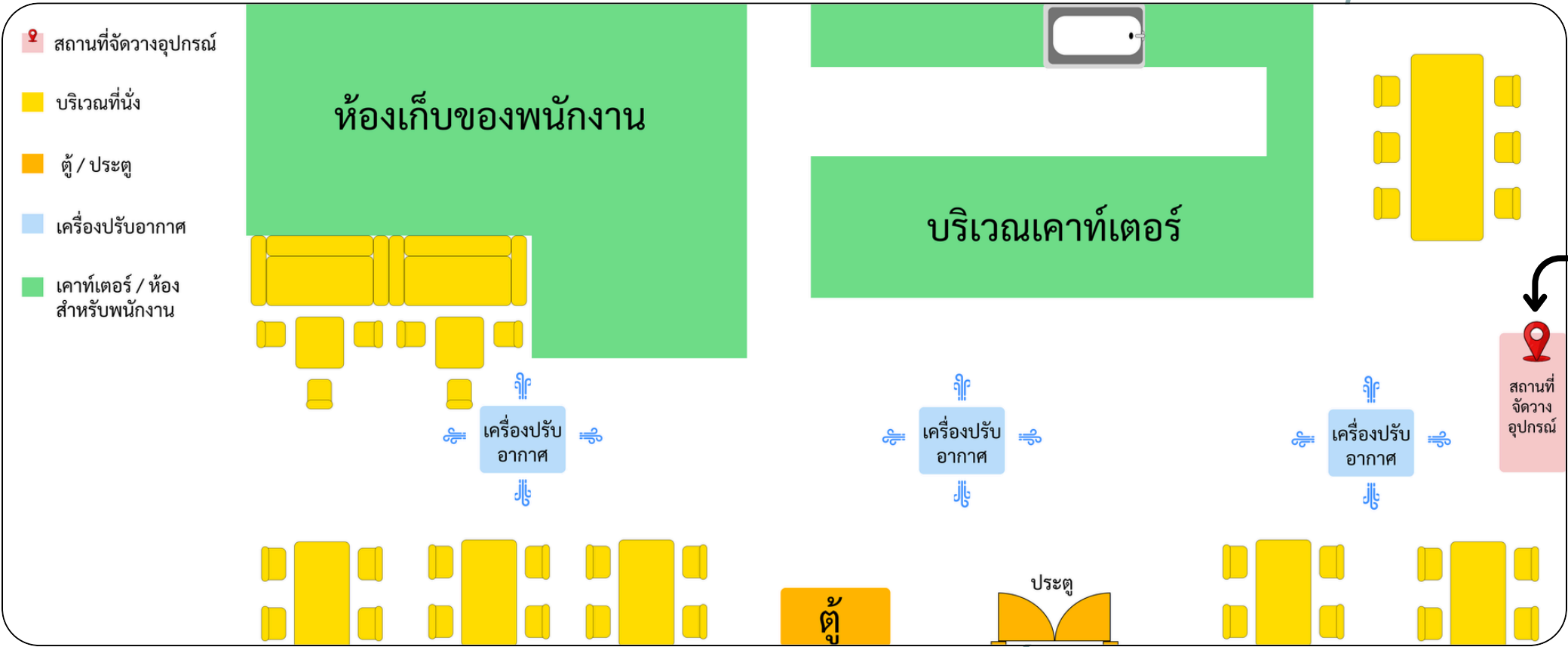


Nova PM & DHT11 Setup

Sensor Location

Location

Cafe Amazon at Lecture Building 6 (ST), Walailak University, Nakron Si Thammarat, Thailand



Sensor Location

Data Collection

PARTICLE COUNT

- PC0.1

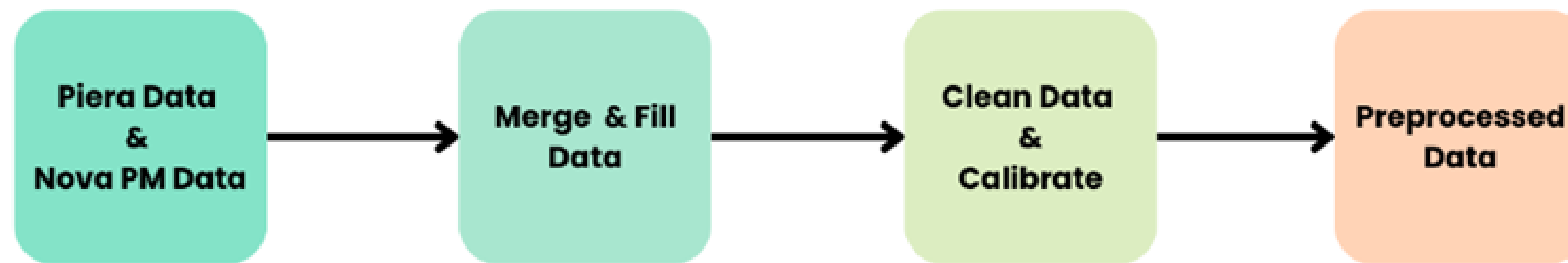
PARTICLE MATTER

- PM2.5, PM10, PM2.5nova, PM10nova

ENVIRONMENT

- Temperature, Humidity, Daytype, Time-range, Frequency

Data Preprocessing

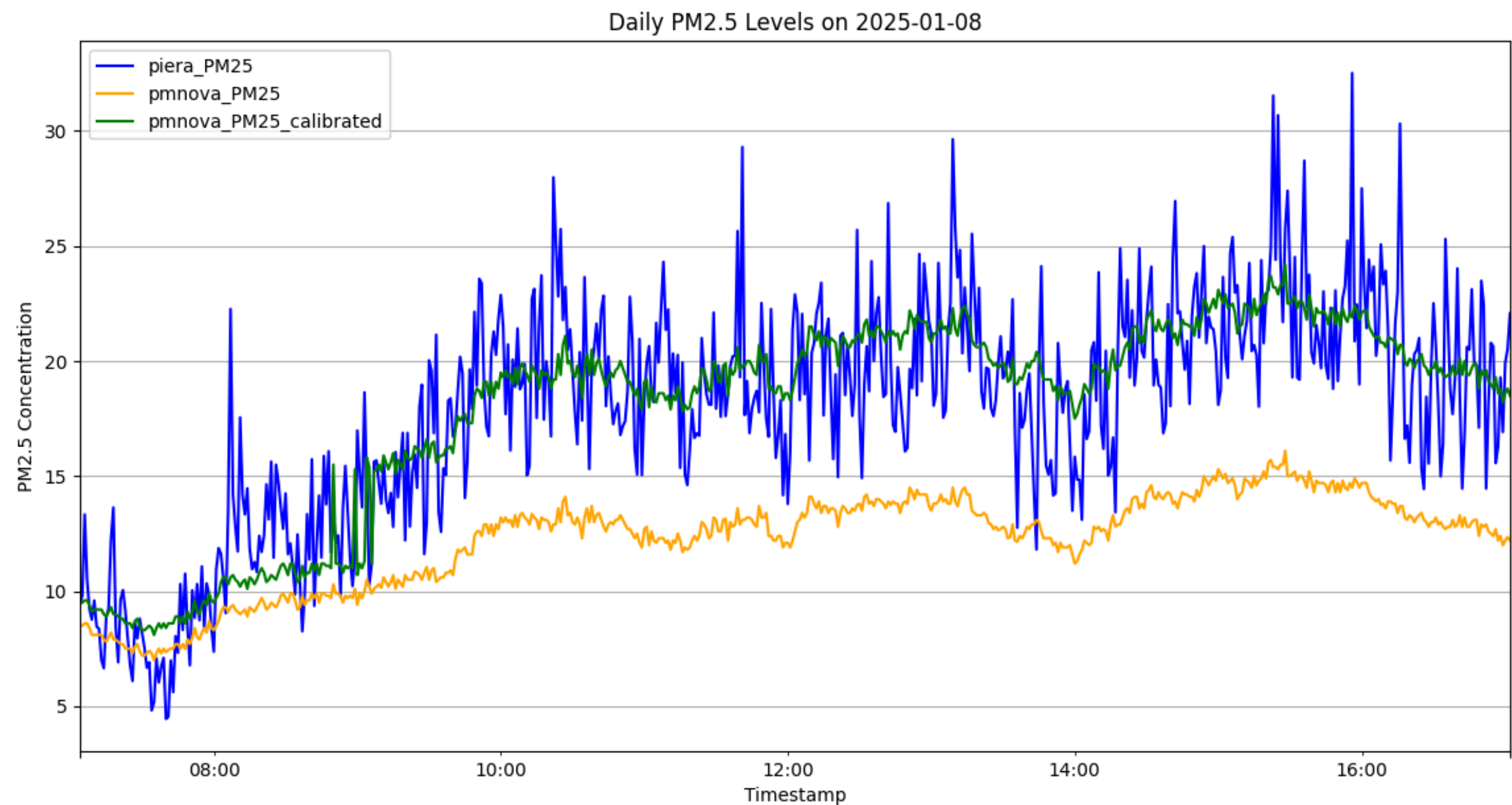


January 6 to June 30, 2025
Total dataset : 980,327 records
5 mins resample dataset : 14,611 records

Library : NumPy, Pandas, Seaborn, Matplotlib Pyplot, Pytorch

Data Preprocessing (cont.)

$$\text{pm25calibrate} = \text{pm25_raw} * \text{range_multiplier} * \text{env_factor} * \text{transparency_boost} + \text{bias}$$



RMSE Before : 4.338 $\mu\text{g}/\text{cm}^3$
RMSE After : 2.721 $\mu\text{g}/\text{cm}^3$

Experiment Setup

In the experiment for predicting ultrafine particulate matter, there will be a total of **2 experiments**

1. Instant Prediction Model

evaluate the feasibility of using PM2.5, PM10, humidity, and temperature as predictors for ultrafine particulate matter (PC0.1), and to determine the most accurate regression model for such prediction.

2. Instant Prediction Model (Using data from other devices for prediction)

test the hypothesis that PM2.5 and PM10 values from other devices can be used to determine PC0.1 values.

Prediction Model – Experiment 1

Objective

This study aims to evaluate the feasibility of using PM2.5, PM10, humidity, and temperature as predictors for ultrafine particulate matter (PC0.1), and to determine the most accurate regression model for such prediction.

Model

RNN (Recurrent Neural Network), LSTM (Long Short-Term Memory), GRU (Gated Recurrent Unit)

Model Evaluation

- R-Square
- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)

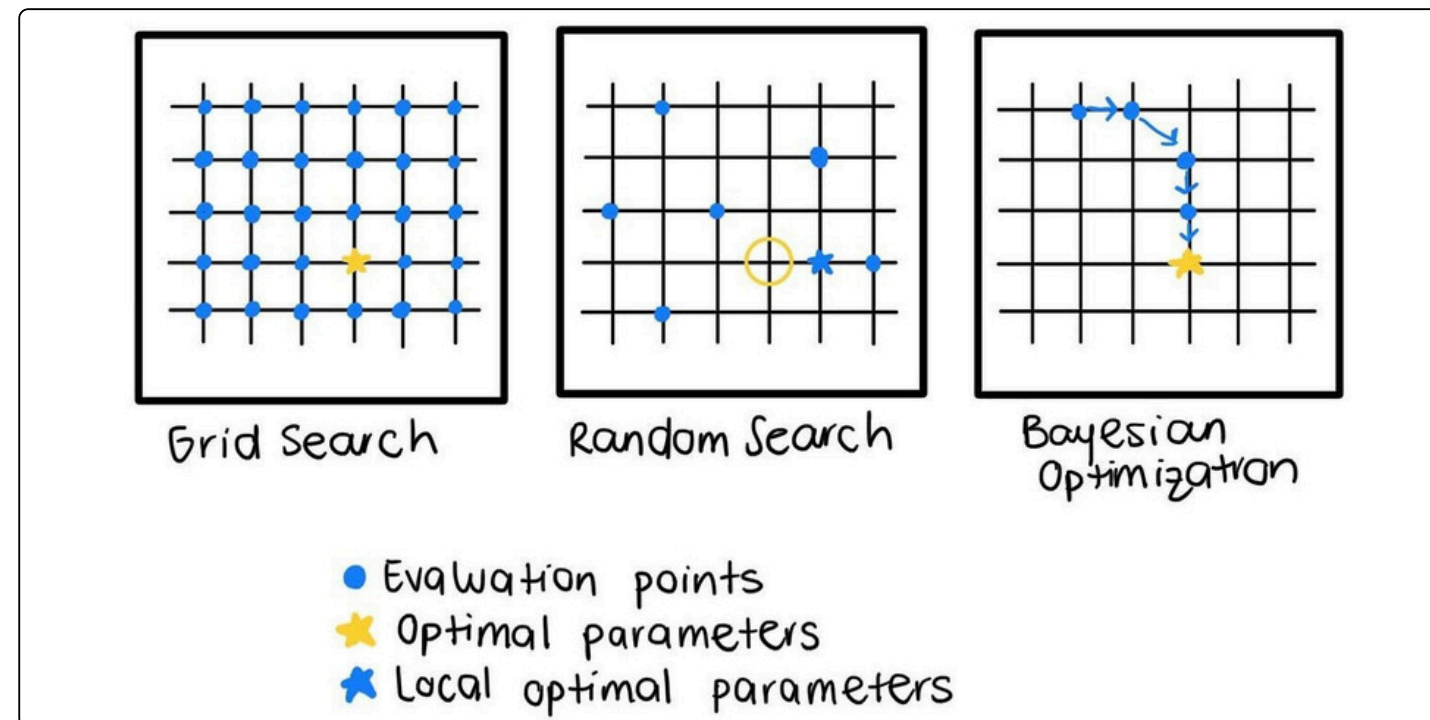
Prediction Model (cont.) – Experiment 1

Hyperparameter Tuning

No.	Feature	Description	Distribution
1	Sequence Length	the number of consecutive elements that an AI model processes in sequential data, such as text, images, or time series.	6–48
2	Hidden Size	a hyper-parameter and it refers to the dimensionality of the vector h_t	32 – 256
3	Number of Layers	the count of processing levels within a technical system, such as a neural network or a network communication protocol stack	1 – 3
4	Dropout Rate	the fraction of neurons that are randomly deactivated during each training iteration	0.1 – 0.5
5	Batch Size	the number of training samples processed at once during a single forward and backward pass of the network	32 – 128
6	Learning Rate	a hyperparameter that controls the size of the step taken during gradient descent to update the model's weights	0.0001 – 0.001
7	Number of Epochs	one complete pass through the entire training dataset once by the neural network	100 – 300
8	Optimizer Type	algorithms responsible for updating the parameters of a neural network during the training process to minimize the loss function	Adam, AdamW, SGD

Prediction Model (cont.) – Experiment 1

Search Space : Bayesian Optimization ($n_{\text{trial}} = 300$)



Data Split : TimeSeriesSplit ($n_{\text{split}} = 5$)

Split 1:	Training set	Test set
Split 2:	Training set	Test set
Split 3:	Training set	Test set
Split 4:	Training set	Test set

Prediction Model (cont.) – Experiment 1

Result

Model	RMSE	MAE	R ²
RNN (sequence_length : 6, hidden_size : 224, num_layers : 3, dropout_rate : 0.432, batch_size : 32, learning_rate : 0.0018)	307.335	151.933	0.557
LSTM (sequence_length: 30, hidden_size : 256, num_layers : 3, dropout_rate : 0.373, batch_size : 32, learning_rate : 0.0019)	353.320	241.150	0.710
GRU (sequence_length : 42, hidden_size : 256, num_layers : 3, dropout_rate : 0.345, batch_size : 32, learning_rate : 0.0035)	264.088	104.486	0.865

**The S.D. of PCO.1 is equal to 851.613 particle/cm³

Prediction Model (cont.) – Experiment 2

Objective

This study aims to test the hypothesis that PM2.5 and PM10 values from other devices can be used to determine PC0.1 values.

Model

GRU (Gated Recurrent Unit)

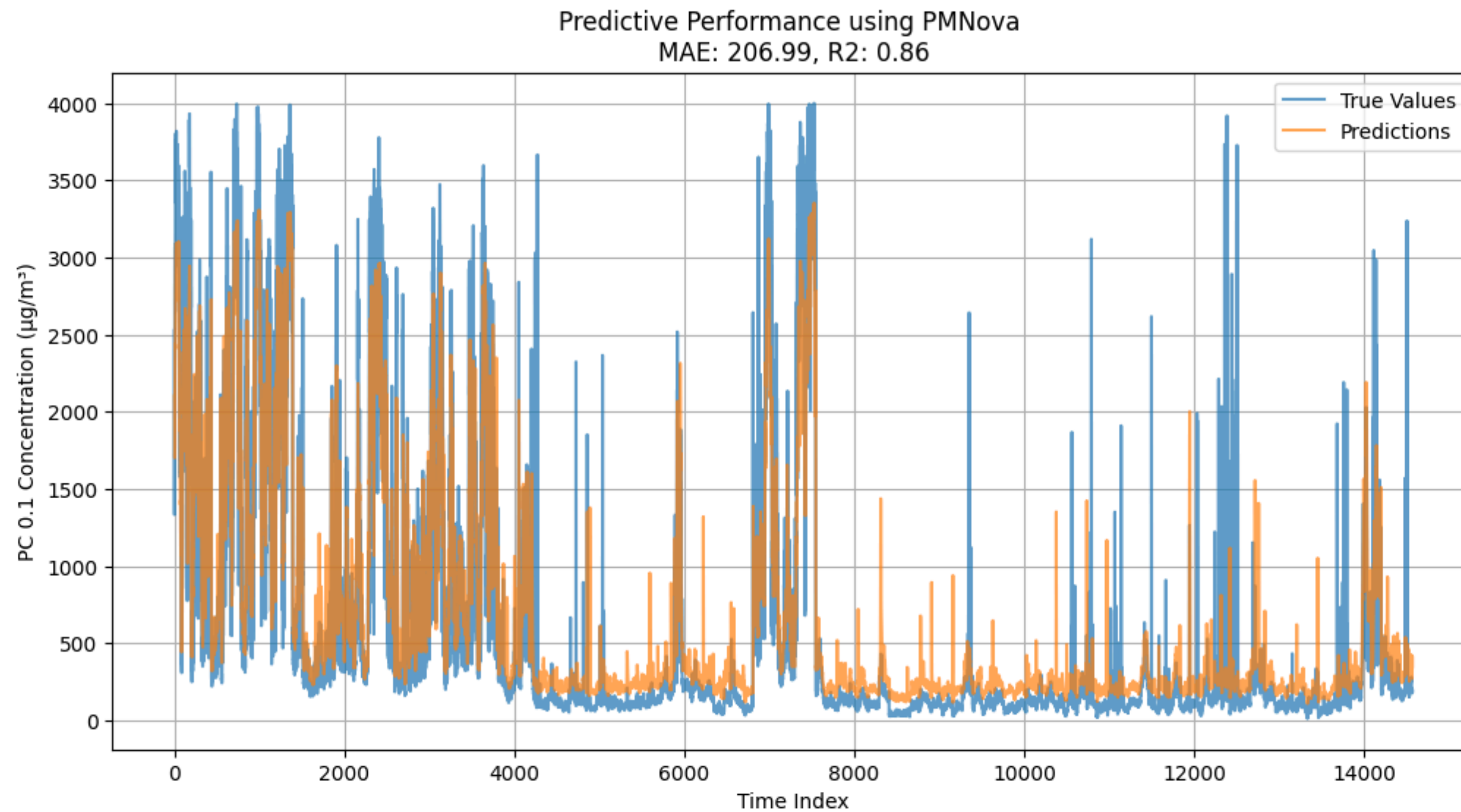
Model Evaluation

- R-Square, Mean Absolute Error (MAE), Root Mean Square Error (RMSE)

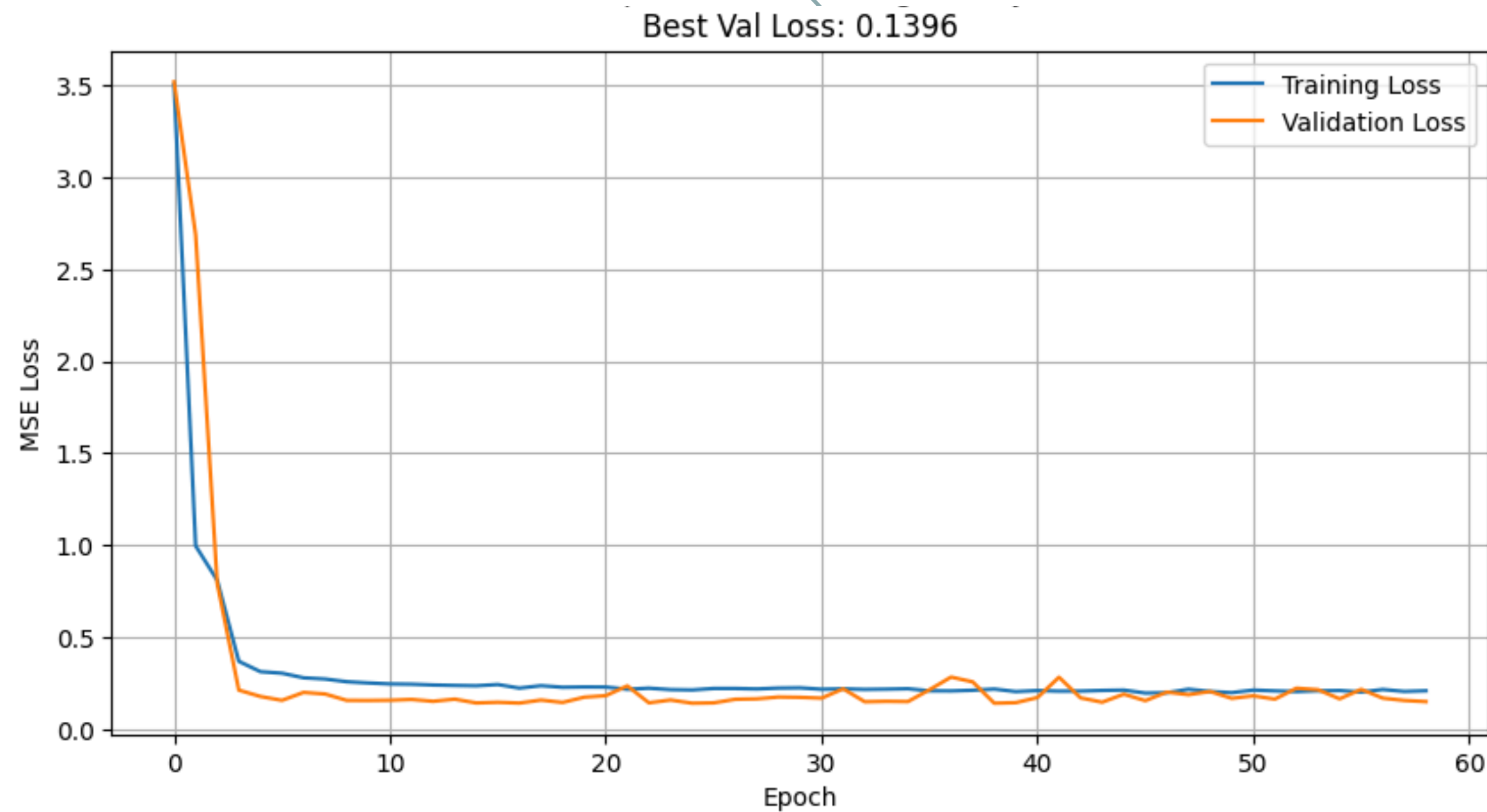
Results

Model	RMSE	MAE	R ²
Piera Device Data	264.088	104.486	0.865
PMNova Device Data	318.680	206.990	0.860

Prediction Model (cont.) – Experiment 2



Prediction Model (cont.) – Experiment 2



Deployment

✓ISTA+ VISTA
Vulnerable Indoor Sensitive Tiny Aerosol monitor

UFPs Monitoring

• LIVE

📍 อาคารวิชาการ 4

08/06/2025

อัปเดต: 15:03:17



🌡️ 27.8 °C 🌧️ 83 %



คลิกที่ Bubble ด้านบนเพื่อดูคำแนะนำสำหรับแต่ละกลุ่ม

ข้อมูลมลพิษทางอากาศ



ระดับคุณภาพอากาศ

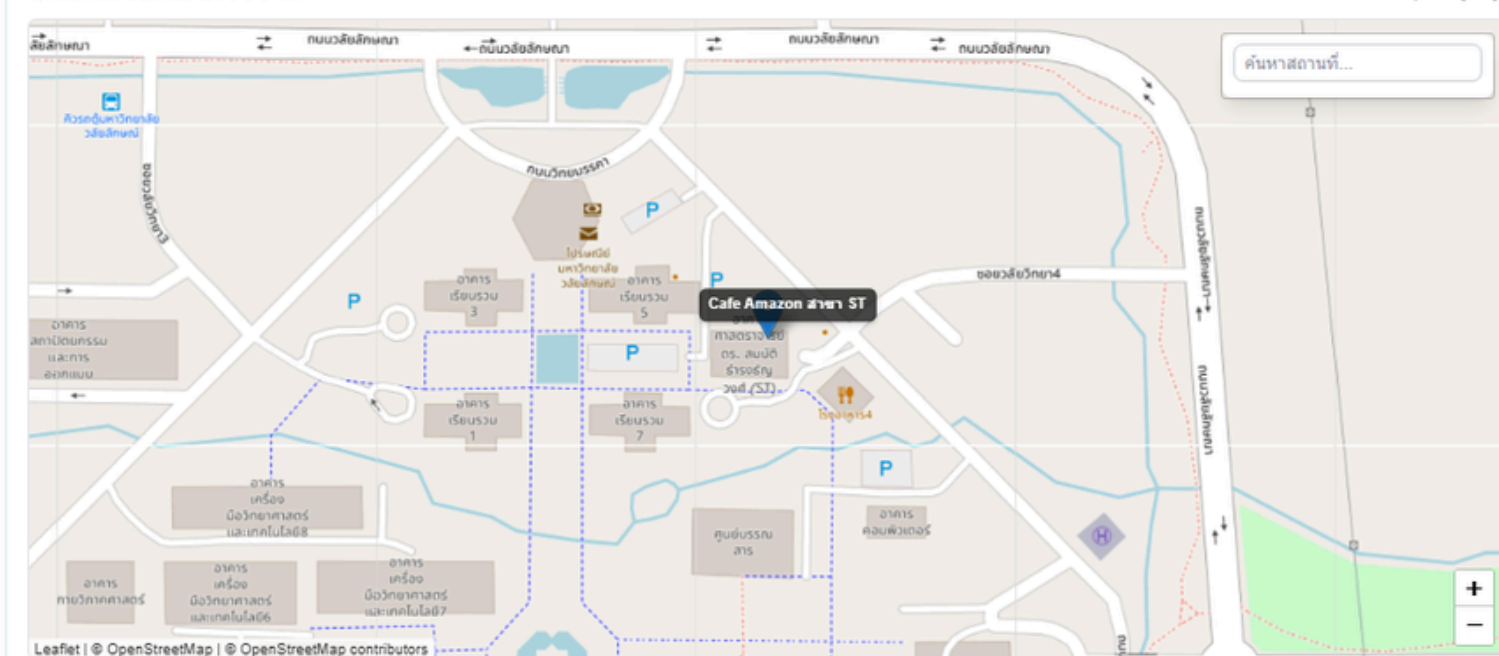


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คลิกเพื่อดูเกร็ดความรู้

Air quality History Data

แผนที่ตำแหน่งตรวจวัด

คลิกที่หมุดเพื่อดูข้อมูล



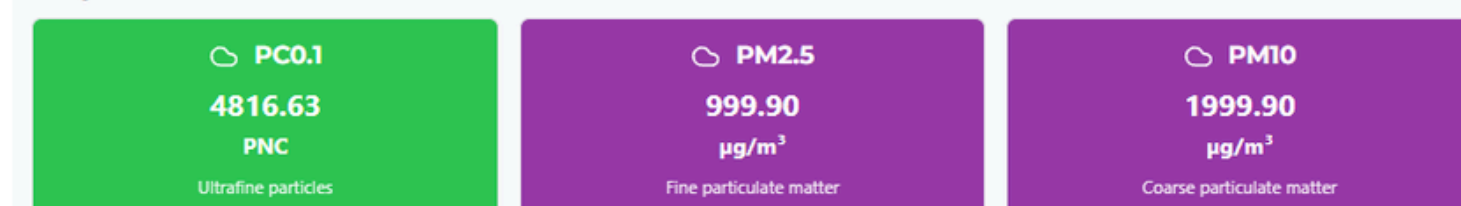
ข้อมูลย้อนหลัง

สถานที่: อาคารวิชาการ 4

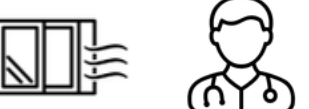
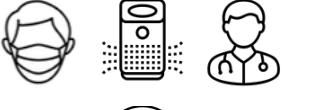
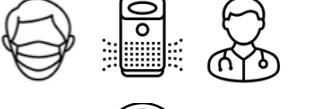
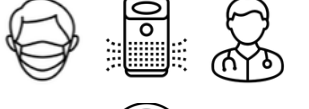
Hourly



Daily

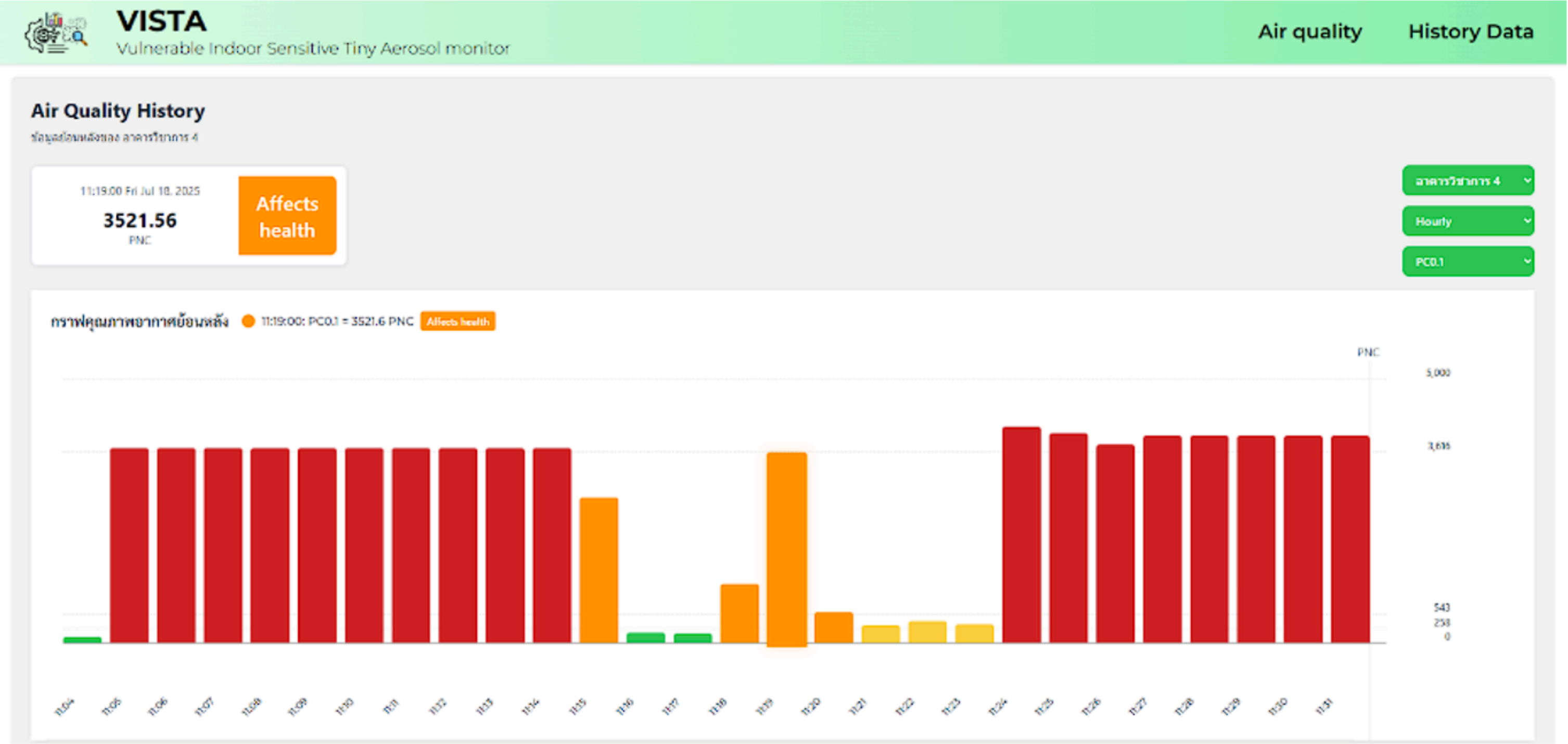


Air Quality Index

Risk Level	Particle count			Particulate Matter Concentration		Health Impact		Health Recommendations	
	PCO.1	Hourly mean PCO.1	Daily mean PCO.1	PM2.5 (µg/m³)	PM10 (µg/m³)	General Public	Sensitive Groups	General Public	Sensitive Groups
Good	0-260	-	1000	0-15.0	0-50.0				
Warning	261-540	-	-	15.1-25.0	51.0-80.0				
Affects Health	541-3620	-	-	25.1-37.5	81.0-120.0				
Danger	3621-6270	-	-	37.6-75.0	121.0-180.0				
Hazardous	6271 and above	20000 and above	10000 and above	75.1 and above	181.0 and above				

*** Currently, Level of PCO.1 mean , PCO.1 daily criteria developed by the World Health Organization and PCO.1 obtained from finding the coefficient between PM2.5 and PM10 ***

Deployment (cont.)



Deployment (cont.)



Operational Problems

Problem and Obstacle	Solution Approach
The Nova SDS011 dust measurement device sends duplicate data for extended periods due to long-term use and the low cost of the equipment.	Regularly maintain and clean the equipment before collecting dust data.
PCO.1 data from the Piera IPS-7100 device cannot be used directly because the data contains significant noise contamination and the measurement volume units are inconsistent.	Data comparison must be performed before use in the Data Preprocessing stage. Calculate in conjunction with PM2.5 dust data to obtain accurate values with appropriate units ready for use in model training.
PM2.5 dust data from the Nova SDS011 device shows significant differences in dust values compared to the Piera IPS-7100 device, making it impossible to use the data for modeling immediately as it would prevent the model from learning dust formation patterns.	Data comparison must be performed before use in the Data Preprocessing stage by calculating in conjunction with Piera IPS-7100 dust data, humidity, and temperature until Nova SDS011 dust values are obtained that are as close as possible to the Piera IPS-7100 device and can be used for model training.
The model has low accuracy because data from each month has different characteristics, and external dust affects indoor dust levels, causing increases.	Collect outdoor dust data and external factors to study daily dust characteristics in conjunction with indoor dust data to improve model accuracy.

Limitations

1. Model Prediction Capabilities

The model cannot predict future ultra-fine particle concentrations.

2. Sensor Hardware Constraints

Nova SDS011 has lower sensitivity than Piera IPS-7100, creating unbridgeable data gaps that render some periods unusable.

3. Prototype System Limitations

Limited accuracy due to technological constraints and insufficient data coverage of dust events.

Suggestions

1.Advanced Model Architectures

Experiment with One-to-Many and Many-to-Many prediction models to identify optimal architecture.

2.Time-Series Forecasting Development

Develop forecasting capabilities using dust data's time dependency characteristics for early warning systems.

3.Enhanced Data Collection

Expand collection periods, integrate environmental factors, and improve sensor calibration methods.

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